

ABBOTSFORD, BC, Canada

WMO: 711080

Lat: 49.027N

Lon: 122.377W

Elev: 194

StdP: 14.59

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	19.7	24.1	1.1	5.9	24.7	7.8	8.3	29.6	28.6	35.1	24.1	38.2	9.3	80	0.516

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	86.2	67.3	82.4	66.1	78.9	64.6	68.9	83.3	66.9	80.2	65.2	77.2	7.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.1	87.1	75.7	61.4	82.0	73.1	60.0	77.7	70.9	33.1	83.8	31.5	80.5	30.1	77.4	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.9	17.5	14.5	DB	15.3	92.8	4.6	3.8	12.0	95.6	9.3	97.8	6.7	100.0	3.4	102.7
			WB	13.4	72.7	4.5	2.5	10.1	74.5	7.5	75.9	4.9	77.3	1.6	79.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.3	38.9	40.8	44.7	49.8	56.1	60.6	65.5	65.7	60.3	51.3	43.5	37.9
	DBStd	11.17	6.73	5.60	5.19	4.94	5.30	4.95	4.65	4.24	4.71	4.73	6.22	6.55
	HDD50	1467	345	258	175	60	8	0	0	0	0	40	206	375
	HDD65	5154	809	677	628	457	280	151	47	42	155	424	645	839
	CDD50	1951	1	1	12	54	197	319	481	486	308	81	10	1
	CDD65	164	0	0	0	1	4	20	63	63	13	0	0	0
	CDH74	2091	0	0	1	14	97	276	758	720	224	1	0	0
	CDH80	598	0	0	0	1	15	83	241	209	49	0	0	0
(14) Wind	WSAvg	5.6	5.9	6.6	6.6	6.0	5.4	5.5	5.2	4.6	4.2	5.0	6.0	6.1
(15) Precipitation	PrecAvg	46.0	5.6	4.1	4.0	3.6	3.2	2.5	1.7	1.9	2.8	4.8	6.1	5.7
	PrecMax	62.5	12.1	10.7	7.8	7.2	7.0	8.3	5.2	6.7	7.3	11.0	11.4	11.1
	PrecMin	28.0	1.2	0.3	0.9	1.2	0.7	0.6	0.0	0.0	0.2	0.6	2.2	0.7
	PrecStd	7.0	2.4	2.3	1.5	1.3	1.5	1.5	1.2	1.6	1.7	2.4	2.1	2.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.3	58.1	66.3	75.5	82.1	87.8	91.9	89.9	85.1	70.9	60.4	56.5
		MCWB	50.7	48.8	53.1	57.9	63.2	67.3	70.4	68.5	65.4	57.8	55.3	52.4
	2%	DB	52.2	53.9	60.2	68.1	76.4	81.8	86.2	85.6	80.1	66.5	56.5	50.9
		MCWB	48.1	45.9	50.2	55.0	60.7	65.0	68.2	67.4	64.3	55.9	52.7	47.5
	5%	DB	49.8	51.3	56.1	63.3	71.7	76.2	81.9	81.6	75.6	62.5	53.9	48.1
		MCWB	46.2	45.3	48.2	52.2	58.9	62.6	66.7	66.0	62.3	54.7	50.4	44.9
	10%	DB	47.4	48.8	53.0	59.1	67.3	71.5	77.7	77.8	71.3	59.3	51.7	45.8
		MCWB	44.4	44.1	46.7	50.1	56.9	60.3	64.6	64.6	60.5	53.5	48.5	43.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.9	51.1	55.0	60.1	64.7	69.3	72.6	70.9	67.9	61.4	56.5	52.5
		MCDB	54.7	54.2	63.7	72.5	78.4	84.6	88.8	85.6	81.4	67.3	58.8	56.1
	2%	WB	49.2	48.5	51.7	56.0	61.9	65.6	69.4	68.4	65.6	58.2	53.6	48.2
		MCDB	51.1	51.5	57.5	65.8	73.7	79.3	83.9	82.7	77.5	62.6	56.0	50.6
	5%	WB	46.7	46.5	49.5	53.4	59.8	63.0	67.2	66.6	63.4	56.0	51.2	45.6
		MCDB	48.7	49.5	54.2	61.0	70.0	74.2	80.2	79.7	73.6	60.6	53.2	47.6
	10%	WB	44.8	44.8	47.7	51.2	57.6	60.9	65.1	64.9	61.3	54.1	49.2	43.5
		MCDB	47.0	48.0	52.1	57.7	66.2	70.6	76.6	76.6	69.1	58.3	51.2	45.3
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	12.0	13.8	15.9	17.3	16.9	20.2	20.7	19.1	14.0	10.8	9.0
		MCDBR	12.6	16.4	19.6	24.6	26.5	27.0	28.8	29.2	28.9	20.2	13.2	11.1
	5% WB	MCWBR	9.0	10.3	11.5	14.0	14.2	13.7	14.3	14.2	15.2	12.1	9.2	8.3
		MCWBR	11.0	13.0	17.0	21.5	23.7	24.5	27.0	26.9	25.8	16.9	11.4	10.5
(40) Clear-Sky Solar Irradiance	taub		0.314	0.309	0.326	0.361	0.388	0.363	0.358	0.361	0.345	0.338	0.319	0.306
		taud	2.457	2.507	2.477	2.376	2.326	2.411	2.445	2.456	2.496	2.502	2.485	2.444
	Ebn at Noon		237	270	283	282	277	285	284	278	272	254	233	223
		Edh at Noon	20	24	29	35	38	36	34	32	28	24	19	18
(44) All-Sky Solar Radiation	RadAvg		284	558	849	1306	1682	1813	2025	1716	1198	661	331	226
	RadStd		43	79	109	132	143	201	169	128	104	86	56	33

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (66 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+36
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page